

Q1.

The function f , where $f(x) = \sec x$, has domain $0 \leq x < \frac{\pi}{2}$ and has inverse function f^{-1} , where
 $f^{-1}(x) = \sec^{-1} x$

- (a) Show that

$$\sec^{-1} x = \cos^{-1} \frac{1}{x} \quad (2)$$

- (b) Hence show that

$$\frac{d}{dx}(\sec^{-1} x) = \frac{1}{\sqrt{x^4 - x^2}} \quad (4)$$

(Total 6 marks)

Q2.

- (a) Given that $u = \sqrt{1-x^2}$, find $\frac{du}{dx}$. (2)

- (b) Use integration by parts to show that

$$\int_0^{\frac{\sqrt{3}}{2}} \sin^{-1} x \, dx = a\sqrt{3}\pi + b$$

where a and b are rational numbers.

(6)

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Q3.

- (a) Differentiate $x \tan^{-1} x$ with respect to x . (2)

- (b) Show that

$$\int_0^1 \tan^{-1} x \, dx = \frac{\pi}{4} - \ln \sqrt{2} \quad (5)$$

(Total 7 marks)